

PAPER_168 — MUGE 3D Simulation Entity Framework: GPU Rendering, Per-System Archives, OpenGL

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \left(1 - [SSq] \cdot U_{b_i} / F_U(r, t)\right), \quad [SSq] = 0.57$$

Abstract

This paper documents the **MUGE 3D Simulation Entity Framework** — the architectural mapping from MUGE physics systems to 3D renderable entities in the Star Magic visualization engine. Each MUGE system (SGR 1745, SgrA*, Tapestry, Westerlund 2, Pillars, Rings, Student's Guide) is assigned a per-system archive directory containing image assets, video clips, and per-system physics plugin DLLs. The rendering system uses OpenGL/GLFW with multi-viewport Camera, procedural terrain generation (Perlin noise + extrusion + boolean operations), and MicroTeX LaTeX math overlay. This provides the Tier 3 VR/VM 3D gateway for the UQFF physics output from Tier 2 calculations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Entity Architecture

1.1 MeshData Struct (OBJ Format)

```
struct MeshData {
    std::vector<glm::vec3>    vertices;    // 3D positions
    std::vector<glm::vec3>    normals;     // Surface normals (physics shading)
    std::vector<glm::vec2>    texCoords;   // UV coordinates
    std::vector<unsigned int> indices;     // Triangle indices
};
```

1.2 SimulationEntity

```
struct SimulationEntity {
    std::string name;           // System name (e.g., "SGR1745")
    std::string archive_path;   // "archive/{name}/"
    std::string image_path;     // "archive/{name}/image.jpg"
    std::string video_path;     // "archive/{name}/video.mp4"
    std::string plugin_path;    // "archive/{name}/plugin.dll"
    MugecSystem* muge_system;   // Pointer to physics system
    MeshData mesh;             // 3D geometry
    glm::vec3 position;         // World position
    glm::quat orientation;      // Orientation quaternion
    float scale;               // Scale factor
};
```

2. populate_simulation_entities()

```
void populate_simulation_entities(
    std::vector<MugeSystem>& muge_systems,
    std::vector<SimulationEntity>& entities
) {
    for (auto& sys : muge_systems) {
        SimulationEntity ent;
        ent.name = sys.name;
        ent.archive_path = "archive/" + sys.name + "/";
        ent.image_path = ent.archive_path + "image.jpg";
        ent.video_path = ent.archive_path + "video.mp4";
        ent.plugin_path = ent.archive_path + "plugin.dll";
        ent.muge_system = &sys;
        ent.position = compute_entity_position(sys);
        ent.orientation = glm::identity<glm::quat>();
        ent.scale = compute_entity_scale(sys);
        // Generate placeholder mesh
        ent.mesh = generateSphereMesh(1.0f, 32, 32);
        entities.push_back(std::move(ent));
    }
}
```

3. Per-System Archive Structure

```
archive/
+-- SGR1745/
|   +-- image.jpg      ? Chandra X-ray image of SGR 1745-2900
|   +-- video.mp4      ? Time-lapse magnetar flare simulation
|   +-- plugin.dll     ? SGR1745 UQFF physics plugin
+-- SgrA/
|   +-- image.jpg      ? EHT Sgr A* shadow image (2022)
|   +-- video.mp4      ? MUGE resonance field animation
|   +-- plugin.dll     ? SgrA* UQFF physics plugin
+-- Tapestry/
|   +-- image.jpg      ? HST Tapestry nebula image
|   +-- video.mp4      ? Star formation field simulation
|   +-- plugin.dll
+-- Westerlund2/
+-- Pillars/
+-- Rings/
+-- StudentGuide/
```

4. OpenGL Rendering Pipeline

4.1 Multi-Viewport Camera System

```
struct Camera {
    glm::vec3 position;
```

```

glm::vec3 target;
glm::vec3 up;
float      fov;           // Field of view [degrees]
float      near_plane;    // Near clipping
float      far_plane;     // Far clipping (set to 1e25 for cosmic scale)

glm::mat4 getViewMatrix() const;
glm::mat4 getProjectionMatrix(float aspect) const;
};

```

Multi-viewport layout: physics view (left) + simulation view (right) + data HUD (top).

4.2 Entity Scale Calculation

Systems span 13 orders of magnitude in size:

System	Physical Size	Entity Scale Factor
SGR 1745	~10 km	0.001
SgrA*	~12 × Rs	1.0 (reference)
Tapestry	~100 ly	100.0
Westerlund2	~400 ly	400.0
Pillars	~4 ly	4.0
Rings	~1 Gly	1000.0
Student	Hubble vol.	1e13

5. Procedural Terrain Generation

```

// Perlin noise heightmap
MeshData generateProceduralLandscape(int width, int height, float scale) {
    MeshData mesh;
    for (int z = 0; z < height; z++) {
        for (int x = 0; x < width; x++) {
            float y = perlinNoise(x * scale, z * scale);
            mesh.vertices.push_back({x * scale, y, z * scale});
        }
    }
    computeNormals(mesh); // Phong shading normals
    computeUVs(mesh);     // UV for physics field texture mapping
    return mesh;
}

// Boolean union (CSG)
MeshData booleanUnion(const MeshData& A, const MeshData& B);

// Extrusion along path
MeshData extrudeMesh(const MeshData& profile, const std::vector<glm::vec3>& path);

Physics fields (MUGE compressed, resonance) are mapped to terrain height via:
 $y_{terrain} = \log_{10}(|g_{MUGE}|) \cdot scale$ 

```

6. MicroTeX LaTeX Overlay

```
// Render physics equations as LaTeX overlay via MicroTeX
void renderLatexOverlay(const std::string& latex_str,
                        float x, float y,
                        float font_size = 16.0f) {
    auto tex_renderer = MicroTeX::create();
    tex_renderer->render(latex_str, x, y, font_size);
}

// Example overlays:
renderLatexOverlay("$g_{comp} = 1.78 \times 10^{39}$", 0.1, 0.9);
renderLatexOverlay("$g_{res} = 1.66 \times 10^{45}$", 0.1, 0.85);
renderLatexOverlay("$F_U = -2.06 \times 10^{59}$", 0.1, 0.80);
```

7. UQFF Integration with Rendering

The simulation loop: 1. Physics compute: `compute_compressed_MUGE()` / `compute_resonance_MUGE()` 2. Field normalization: $g \propto \log_{10}(|g|)$ for visual range compression 3. Terrain update: heightmap updated from new MUGE values each frame 4. Particle update: `addUQFFBodyForce()` applied to fluid particles 5. LaTeX HUD: Current MUGE values rendered as math equations in overlay 6. Entity update: position + orientation updated from time-evolved UQFF

8. CP/Architecture Integration

This framework resides in **Tier 3 VR/VM** (`source2(HEAD PROGRAM).cpp`, ~2,625 lines, GPU-heavy). Connection to CP calculators: - CP1/CP2 compute MUGE values $\propto$ stored in `uqff_results.json` - `populate_simulation_entities()` reads `uqff_results.json` $\propto$ creates 3D entities - Source2 GUI (Tab for 3D Viewer) $\propto$ renders entities via OpenGL pipeline - Per-system plugin DLLs can call CP2/CP3 calculators at runtime

Status: $\propto$ Complete | **CP Stage:** Architectural/Tier 3 VR/VM **Supersedes:** N/A (new framework) | **Related:** PAPER_072 (source2 GUI arch), PAPER_157 (Solar System entity params), PAPER_168 connects to CP2/CP3 via `uqff_results.json`